

# Chemical bonding & hybridisation — Issue #33 candidate:

## Source questions

**Q1** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Electron-domain geometry versus molecular shape

**Concept difficulty:** Medium concept

**Question difficulty:** Connected reasoning

**Family:** Domain count before hybrid label

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

BF<sub>3</sub> and NH<sub>3</sub> each have three sigma bonds to the central atom. BF<sub>3</sub> has no central lone pair; NH<sub>3</sub> has one. Explain why three

### Representation

sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

### Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route • open if you want a warning**

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Supplied three-bond comparison



Connectivity layout only; page angles are not measured ge  
Supplied situation

**Need a hint? • guided support**

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Need the concept again?

Electron-domain geometry versus molecular shape

**Q2** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Sigma/pi orbital inventory without double-counting

**Concept difficulty:** Hard concept

**Question difficulty:** Connected reasoning

**Family:** Sigma/pi orbital allocation

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

For  $\text{H}_2\text{C}=\text{O}$ , assign the carbon hybridisation, count the sigma and pi components of the  $\text{C}=\text{O}$  bond, and identify the carbon orbital left available for the pi component. Keep electron-domain counting separate from counting shared electron pairs.

### Representation

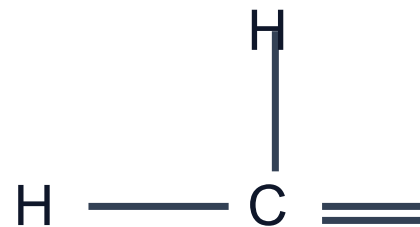
## Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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H<sub>2</sub>C=O connectivity



H<sub>2</sub>C=O connectivity

Connectivity layout only; page angles are not measured

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Sigma/pi orbital inventory without double-counting

**Q3** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Sigma/pi orbital inventory without double-counting

**Concept difficulty:** Hard concept

**Question difficulty:** Connected reasoning

**Family:** Sigma/pi orbital allocation

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

Ethene is  $\text{H}_2\text{C}=\text{CH}_2$ . Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

### Representation

### Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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## Ethene Lewis connectivity



Ethene connectivity

Connectivity layout only; page angles are not measured

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Sigma/pi orbital inventory without double-counting

### COMPETITION TRANSFER

#### Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

**Q4** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry **Topic:** Chemical bonding and hybridisation

**Concept:** Sigma/pi orbital inventory without double-counting **Concept difficulty:** Hard concept

**Question difficulty:** Connected reasoning **Family:** Sigma/pi orbital allocation **Question type:** Constructed response

**Source:** Owner-supplied question **Provenance:** Owner-supplied question

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### ATTEMPT FIRST

For CO<sub>2</sub>, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

#### Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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### Representation

#### CO<sub>2</sub> Lewis connectivity



#### CO<sub>2</sub> connectivity

Connectivity layout only; page angles are not measured

Supplied situation

**Need a hint? · guided support**

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**Need the concept again?**

Sigma/pi orbital inventory without double-counting

**Q5** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Domain count before hybridisation label

**Concept difficulty:** Medium concept

**Question difficulty:** Connected reasoning

**Family:** Domain count before hybrid label

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

Consider the claim: 'Any central atom with three sigma bonds must be  $sp^2$ .' Given  $BF_3$ : three B–F sigma bonds, zero boron lone pairs, planar;  $NH_3$ : three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations

### Representation

support the claim, give the decisive warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

### Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route • open if you want a warning**

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Supplied BF<sub>3</sub> and NH<sub>3</sub> counterexample candidate



BF<sub>3</sub> Lewis inventory

NH<sub>3</sub> Lewis

Connectivity layout only; page angles are not measured geometrically  
Supplied situation

**Need a hint? • guided support**

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Need the concept again?

Domain count before hybridisation label

**Q6** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Domain count before hybridisation label

**Concept difficulty:** Medium concept

**Question difficulty:** Direct application

**Family:** Domain count before hybrid label

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

In CH<sub>3</sub>F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH<sub>4</sub> even though the attached atoms are different. Do not assume all bond properties are identical.

### Representation

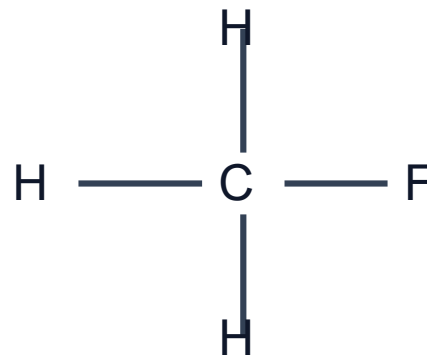
## Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

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## CH<sub>3</sub>F and CH<sub>4</sub> connectivity



CH<sub>3</sub>F connectivity



CH<sub>4</sub> con

Connectivity layout only; page angles are not measured ge  
Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Domain count before hybridisation label

**Q7** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Sigma/pi orbital inventory without double-counting

**Concept difficulty:** Hard concept

**Question difficulty:** Connected reasoning

**Family:** Sigma/pi orbital allocation

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

HCN is represented as  $\text{H}-\text{C}\equiv\text{N}$  with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the  $\text{C}\equiv\text{N}$  bond. Explain how the local domain count is compatible with those components.

### Representation

## Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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## Supplied HCN Lewis connectivity



HCN connectivity; : is the 1

Connectivity layout only; page angles are not meas

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Sigma/pi orbital inventory without double-counting

**Q8** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Electron-domain geometry versus molecular shape

**Concept difficulty:** Medium concept

**Question difficulty:** Connected reasoning

**Family:** Electron-domain geometry versus molecular shape

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

For methylamine,  $\text{CH}_3\text{-NH}_2$ , carbon has four single bonds and nitrogen has three single bonds plus a lone pair. Compare the introductory local hybridisation and arrangement of bonded atoms at carbon and nitrogen. Explain why the same

### Representation

hybridisation label does not require identical molecular shapes.

### Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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## Supplied methylamine connectivity



Nitrogen lone pair is n

Connectivity layout only; page angles are not measured

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Electron-domain geometry versus molecular shape

**Q9** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry

**Topic:** Chemical bonding and hybridisation

**Concept:** Sigma/pi orbital inventory without double-counting

**Concept difficulty:** Hard concept

**Question difficulty:** Connected reasoning

**Family:** Sigma/pi orbital allocation

**Question type:** Constructed response

**Source:** Owner-supplied question

**Provenance:** Owner-supplied question

### ATTEMPT FIRST

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

### Representation

## Conditions

- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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## Supplied local carbon jobs

Three sigma directions; no lone pair

One pi participation

Available valence basis: one 2s and three

Connectivity layout only; page angles are not measured

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Sigma/pi orbital inventory without double-counting

### COMPETITION TRANSFER

#### Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

**Q10** Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

**Subject:** Chemistry **Topic:** Chemical bonding and hybridisation

**Concept:** Electron-domain geometry versus molecular shape **Concept difficulty:** Medium concept

**Question difficulty:** Connected reasoning **Family:** Electron-domain geometry versus molecular shape

**Question type:** Constructed response **Source:** Owner-supplied question **Provenance:** Owner-supplied question

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### ATTEMPT FIRST

A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.'

Evaluate the statement using CH<sub>4</sub>, NH<sub>3</sub> and H<sub>2</sub>O, with their respective central lone-pair counts of zero, one and two.

#### Conditions

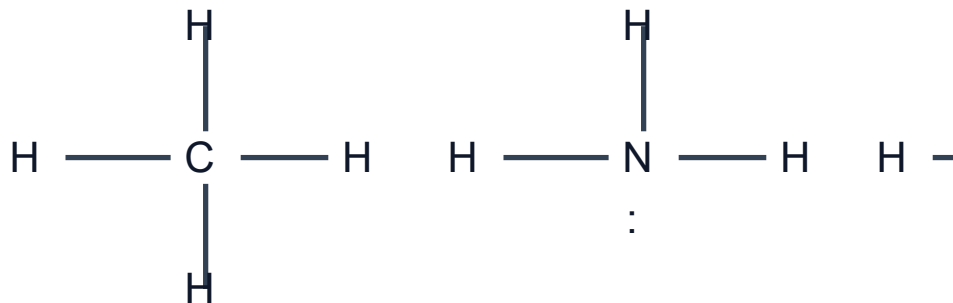
- Use the bounded introductory localized hybrid-orbital model requested by the question; do not treat molecular shape alone as direct measurement of orbital mixing.

**Common wrong route · open if you want a warning**

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#### Representation

Supplied four-domain comparisons



CH<sub>4</sub> connectivity

NH<sub>3</sub> Lewis inventory

H<sub>2</sub>O I

Connectivity layout only; page angles are not measured geometry.

Supplied situation

**Need a hint? · guided support**

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Need the concept again?

Electron-domain geometry versus molecular shape

### **COMPETITION TRANSFER**

#### **Hold domain geometry fixed while changing atom occupancy**

Original transfer task; not claimed as an official exam/PYQ item.

A four-domain centre is changed from four bonds to two bonds plus two lone pairs without changing the total domain count. Explain which geometry description is preserved and which atom-only description must be reconsidered.